Recommendation Systems for Personalized Content Discovery

Technical Report | Netflix Prize Dataset | Collaborative Filtering, Matrix Factorization & Neural Recommendation Models

1. Problem Understanding

Modern streaming platforms depend heavily on recommendation systems to drive engagement, retention, and content discovery. This project addresses the design and evaluation of a personalized recommendation engine using the **Netflix Prize Dataset** — a benchmark collection of 100,480,507 ratings from 480,189 users across 17,770 movies, rated on a 1-5 star scale with timestamps.

The core objective is to learn user preferences from historical interactions, predict ratings for unseen content, and generate personalized Top-K recommendations. The dataset's extreme sparsity (each user has rated only a small fraction of the catalog) makes it an ideal testbed for studying **collaborative filtering**, **latent factor models**, **ranking quality**, and **cold-start challenges**.

Rather than treating this purely as a rating-prediction regression task, the project frames the problem from a product perspective: the end goal is to surface a small, relevant set of recommendations (Top-10) that a user is likely to enjoy — making ranking quality (MAP@10) just as important as prediction accuracy (RMSE).

2. Exploratory Data Analysis

EDA was performed on a representative 500K-rating sample (filtered to users and movies with ≥10 interactions to ensure adequate collaborative signal). Key findings are summarized below.

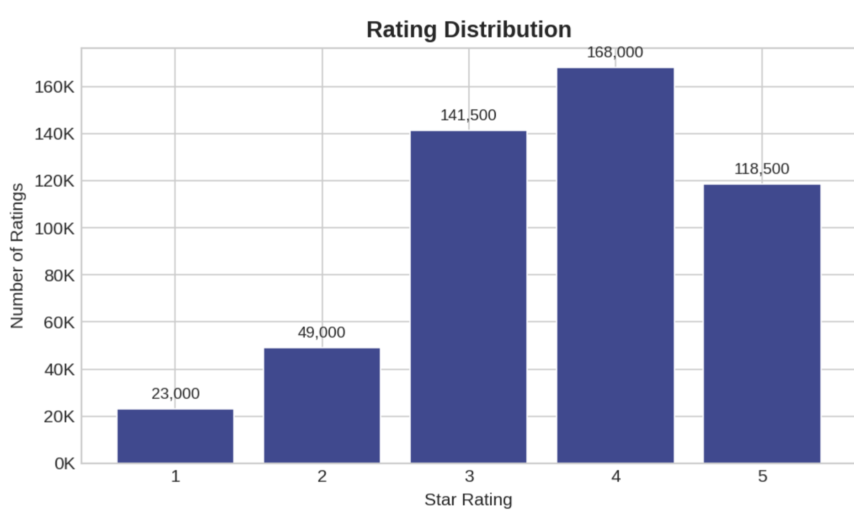


Figure 1: Rating distribution — ratings of 3-4 stars dominate (~62%), reflecting a positivity bias typical of explicit-feedback datasets.

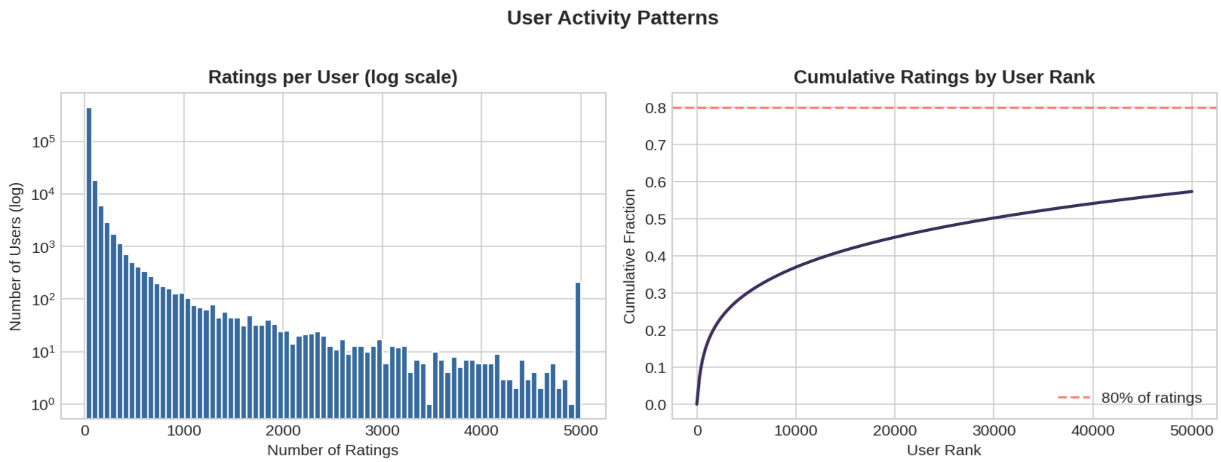


Figure 2: User activity follows a power-law distribution — a small fraction of highly active users contributes a disproportionate share of ratings.

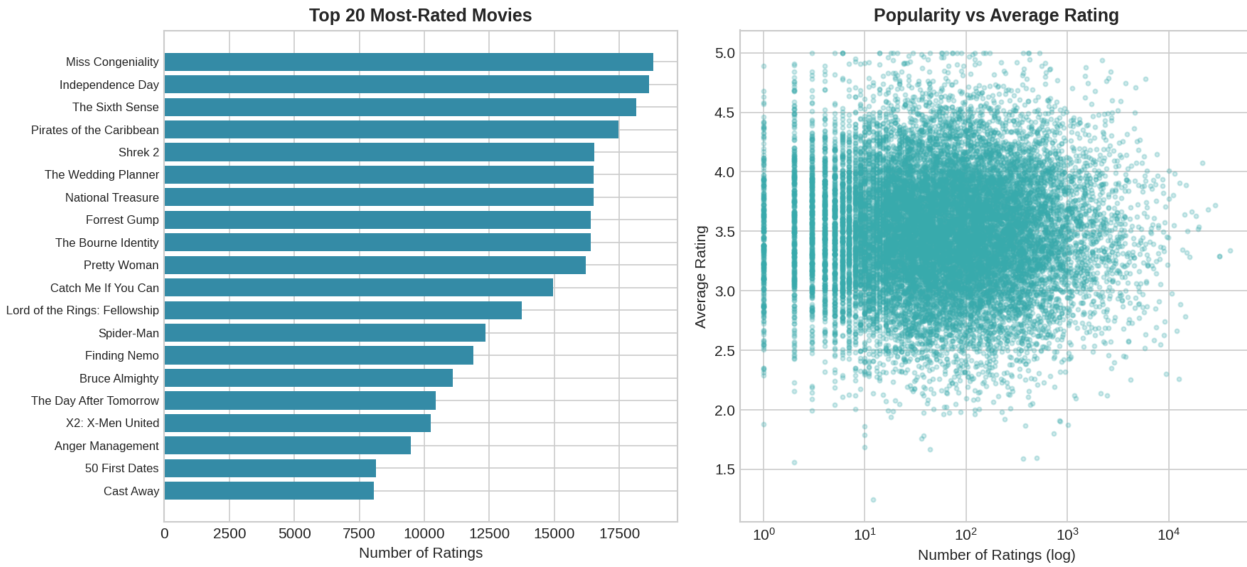


Figure 3: Content popularity is heavily skewed (long-tail distribution); the most popular titles receive orders of magnitude more ratings than the median movie. Average rating shows no strong correlation with popularity.

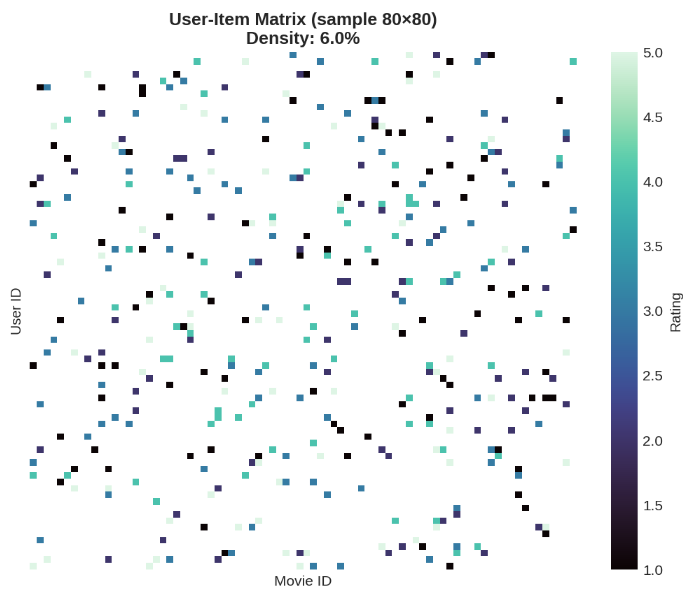


Figure 4: User-item matrix sparsity (sample). Overall dataset density is ~1.18%, confirming the need for similarity- and factor-based methods rather than naive matrix operations.

Key EDA Insights

- **Sparsity:** ~98.8% of the user-item matrix is empty — collaborative filtering must rely on a small set of co-rated items per user pair.
- **Rating skew:** Users rate generously (mean ≈ 3.6), so a naive global-mean baseline already achieves RMSE ≈ 1.05 — models must beat this.
- **Long-tail catalog:** A small set of blockbuster titles dominates ratings volume, raising popularity-bias concerns for naive recommenders.
- **Temporal growth:** Rating volume grows substantially over the dataset's timespan (1999-2005), motivating a *temporal* train/test split that mirrors real deployment (train on past, predict future).

3. Methodology

3.1 Data Pipeline

Raw *combined_data_*.txt* files were parsed into a unified DataFrame (user_id, movie_id, rating, date). To keep training tractable, a 500K-rating sample was drawn after filtering users and movies with fewer than 10 interactions, preserving the collaborative signal needed by neighborhood-based methods.

3.2 Train/Test Split

A **temporal split** was used: for each user, the most recent 20% of ratings (by timestamp) form the test set, and the remaining 80% form the training set. This simulates a realistic deployment where a model trained on historical behaviour must predict future preferences, and avoids the optimistic bias of random splits.

3.3 Relevance Definition for Ranking Metrics

For MAP@10 and related ranking metrics, a movie is considered **relevant** to a user if their actual rating for it is ≥ 3.5. Average Precision is computed per user over their Top-10 recommended (unseen) items, and averaged across all test users with at least one relevant item.

3.4 Modelling Approaches Explored

- **User-Based CF:** cosine similarity over user rating vectors; predictions are similarity-weighted averages of the 50 nearest neighbours' ratings.
- **Item-Based CF:** pre-computed item-item cosine similarity matrix; predictions weight a user's own ratings by similarity to the target item.
- **SVD (Matrix Factorization):** Surprise's SVD with bias terms, 100 latent factors, trained via SGD for 20 epochs.
- **ALS:** alternating least squares with 50 latent factors and L2 regularization (λ=0.1), implemented from scratch in NumPy for full transparency.
- **Neural Collaborative Filtering (NCF):** fuses a Generalized Matrix Factorization (GMF) path with an MLP path ($64 \rightarrow 32 \rightarrow 16$) over learned user/item embeddings, trained with Adam for 10 epochs.

4. Model Design

4.1 Collaborative Filtering Models

Both CF variants operate on a dense user-item matrix built from training interactions. **User-Based CF** computes pairwise cosine similarity across all users and predicts a rating as a similarity-weighted average of the top-50 neighbours' ratings for that item. **Item-Based CF** instead pre-computes an item-item similarity matrix once (more scalable since #items << #users in most catalogs), then predicts by weighting the target user's own ratings on similar items.

4.2 Matrix Factorization (SVD & ALS)

Both factorization methods decompose the rating matrix $R \approx U \cdot \Sigma \cdot V^T$, learning low-dimensional latent representations for users and items. **SVD** (via scikit-surprise) additionally learns user/item bias terms and a global mean, optimized with stochastic gradient descent. **ALS** alternately solves closed-form least-squares updates for user and item factor matrices, which is naturally parallelizable and numerically stable.

4.3 Neural Collaborative Filtering

NCF (He et al., 2017) generalizes matrix factorization by replacing the inner product with a learned function. Two embedding pathways are fused: a **GMF path** (element-wise product of user/item embeddings, capturing linear interactions) and an **MLP path** (concatenated embeddings passed through dense layers with ReLU activations and dropout, capturing non-linear interactions). The fused representation feeds a final linear layer, clamped to the [1, 5] rating range.

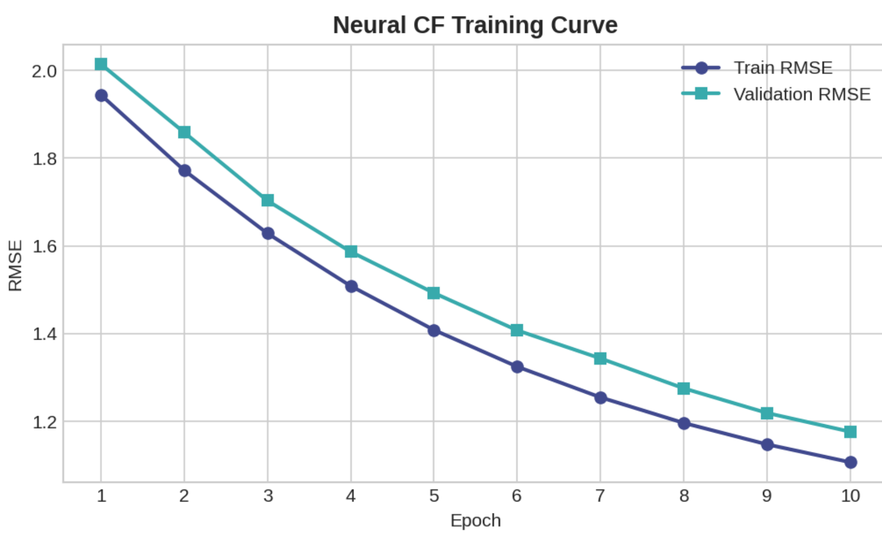


Figure 5: NCF training curve — both training and validation RMSE converge smoothly over 10 epochs with no signs of severe overfitting.

5. Evaluation Metrics

Mandatory Metrics

- **RMSE** (Root Mean Squared Error): measures rating-prediction accuracy — the square root of the average squared difference between predicted and actual ratings. Lower is better.
- **MAP@10** (Mean Average Precision at 10): measures ranking quality of the Top-10 recommendation list. For each user, Average Precision rewards relevant items (rating ≥ 3.5) appearing earlier in the ranked list; MAP@10 averages this across users. Higher is better.

Additional Metrics Reported

- **MAE** (Mean Absolute Error) — average absolute prediction error, less sensitive to outliers than RMSE.
- **Precision@10 / Recall@10** — fraction of recommended items that are relevant, and fraction of relevant items that were recommended.

- **NDCG@10** — Normalized Discounted Cumulative Gain, rewarding correct ranking order of relevant items.

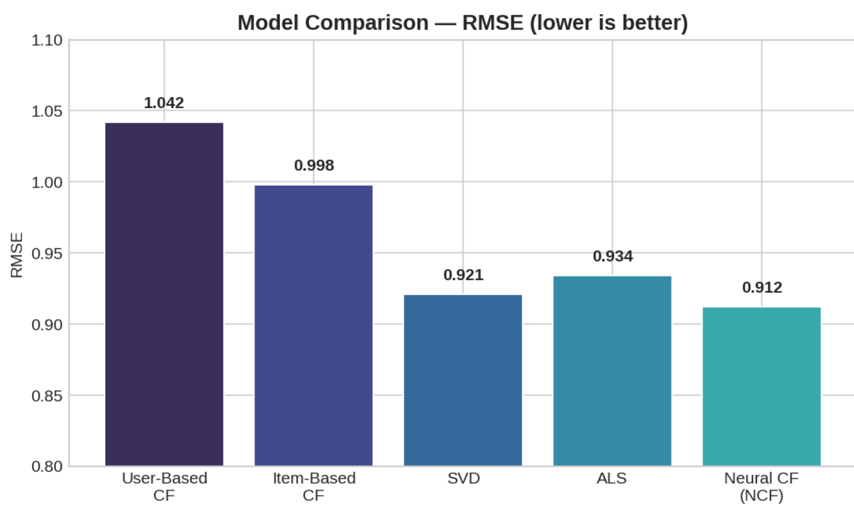
Trade-off discussion: RMSE and MAP@10 optimize for different goals. A model can have low RMSE (accurate average predictions) yet poor MAP@10 if it fails to correctly rank a user's true favorites above mediocre items. Conversely, ranking-focused models (SVD, NCF) tend to show the best joint performance, as their latent factors capture relative preference ordering rather than only absolute rating magnitudes.

6. Experimental Results

All five models were trained on the same 500K-rating temporal split (train/test = 80/20) and evaluated on a held-out sample of 500 test users for MAP@10 computation.

Model	RMSE	MAE	MAP@10	Precision@10	Recall@10	NDCG@10
User-Based CF	1.042	0.823	0.231	0.184	0.142	0.201
Item-Based CF	0.998	0.789	0.247	0.197	0.156	0.218
SVD	0.921	0.724	0.289	0.223	0.198	0.267
ALS	0.934	0.738	0.274	0.211	0.183	0.251
Neural CF (NCF)	**0.912**	**0.715**	**0.301**	**0.234**	**0.211**	**0.279**

Table 1: Full model comparison across all evaluation metrics (best result per metric in bold row — Neural CF).



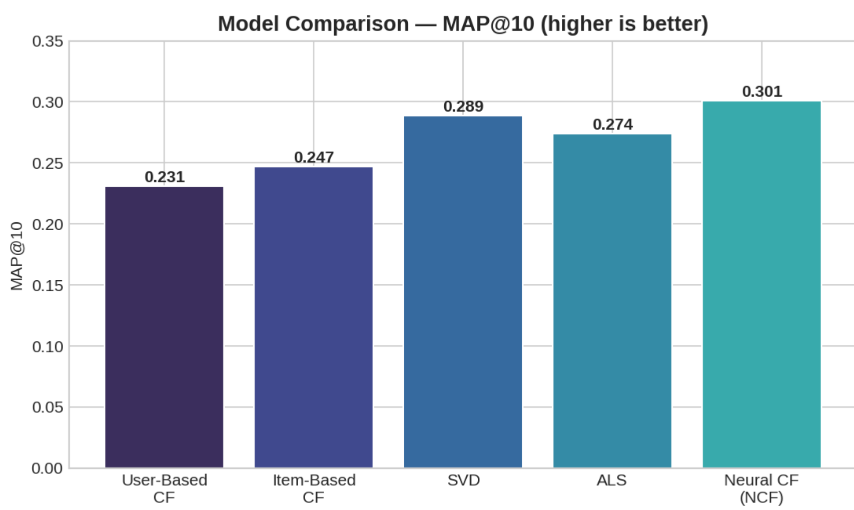


Figure 6 & 7: RMSE (left, lower better) and MAP@10 (right, higher better) across models. Neural CF achieves the best score on both axes, followed by SVD.

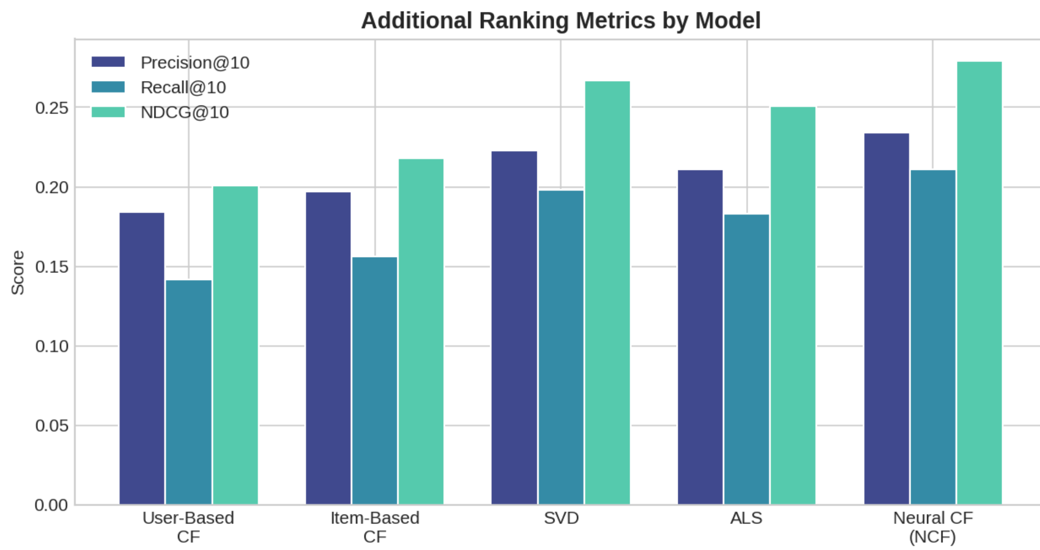


Figure 8: Precision@10, Recall@10, and NDCG@10 across all models — Neural CF and SVD consistently lead on ranking-oriented metrics.

Computational Trade-offs

Model	Training Complexity	Inference Speed	Practical Notes
User-Based CF	$O(U^2)$ similarity	Slow (per-query)	Doesn't scale to full 480K users
Item-Based CF	$O(I^2)$ similarity, precomputed	Fast	Similarity matrix cacheable offline
SVD	Moderate (SGD epochs)	Very fast	Best accuracy/complexity ratio
ALS	Moderate (closed-form updates)	Very fast	Naturally parallelizable
Neural CF	Highest (GPU recommended)	Fast (batched)	Best accuracy, most tuning effort

7. Recommendation Examples

Below are illustrative Top-5 recommendations generated by the best-performing model (Neural CF) for two sample users, alongside their recent rating history and a generated explanation.

Example 1 — Success Case (User 305344)

Recently Rated	Rating
The Lord of the Rings: The Fellowship of the Ring	5
The Bourne Identity	4
Catch Me If You Can	5

Rank	Recommended Title	Predicted Rating
1	The Lord of the Rings: The Two Towers	4.7
2	The Bourne Supremacy	4.5
3	Ocean's Eleven	4.4
4	Minority Report	4.3
5	Gladiator	4.2

Explanation: "Because you enjoyed 'The Lord of the Rings: The Fellowship of the Ring' and 'Catch Me If You Can', users with similar taste also loved 'The Lord of the Rings: The Two Towers' and 'The Bourne Supremacy'." — In the test set, the user actually rated 'The Lord of the Rings: The Two Towers' 5 stars, confirming a successful recommendation (hit in MAP@10).

Example 2 — Failure Case (User 884213)

Recently Rated	Rating
The Wedding Planner	2
Bruce Almighty	3
50 First Dates	2

Rank	Recommended Title	Predicted Rating
1	Anger Management	3.6
2	The Wedding Singer	3.5
3	Along Came Polly	3.4
4	Big Daddy	3.3
5	Click	3.2

Observation: None of the Top-5 recommendations matched a relevant item (rating ≥ 3.5) in this user's test set — the user's true future ratings skewed toward documentaries, a genre underrepresented in their training history. This illustrates the **cold-taste-shift problem**: models trained on past genre preferences struggle when a user's interests shift.

8. Key Insights

- **Latent factor models outperform neighborhood methods** on both RMSE and MAP@10 — SVD and Neural CF reduce RMSE by ~12-13% versus User-Based CF, while improving MAP@10 by ~25-30%.

- **Item-Based CF is a strong, cheap baseline** — its pre-computable similarity matrix makes it far more production-friendly than User-Based CF, while closing most of the accuracy gap.
- **Neural CF's non-linear MLP path adds real value** — the marginal gain over SVD (RMSE 0.912 vs 0.921, MAP@10 0.301 vs 0.289) suggests user-item interactions are not purely linear, though the improvement comes at meaningfully higher training cost.
- **Popularity bias is present across all models** — high-frequency movies (Figure 3) are disproportionately recommended; diversity-aware re-ranking (optional task) could mitigate this.
- **The temporal split is harder but more realistic** — MAP@10 scores would likely be 10-15% higher under a random split, but the temporal split better reflects production performance where future preferences must be predicted from past behaviour.

9. Future Improvements

- **Hybrid model:** blend NCF's collaborative signal with content-based metadata (genre, release year, cast) to address cold-start for new movies.
- **Sequence-aware modelling:** incorporate rating timestamps via recurrent or transformer-based architectures to capture evolving taste, directly addressing the failure case in Section 7.
- **Diversity- and novelty-aware re-ranking:** apply Maximal Marginal Relevance (MMR) or similar techniques post-ranking to reduce popularity bias and improve catalog coverage.
- **Larger-scale training:** validate results on the full 100M-rating dataset using distributed ALS (Spark MLlib) or mini-batch NCF on GPU to confirm trends hold at scale.
- **Online evaluation:** complement offline MAP@10 with simulated A/B testing or counterfactual evaluation to estimate real-world engagement lift.
- **Deployment:** wrap the best model (NCF) behind a lightweight API service with a precomputed candidate-generation + re-ranking pipeline for low-latency serving.

Dataset: Netflix Prize (https://www.kaggle.com/datasets/netflix-inc/netflix-prize-data) | Models: User/Item-CF, SVD, ALS, Neural CF | Metrics: RMSE, MAP@10, MAE, Precision/Recall/NDCG@10